

symcoder Phase-3 Vandermonde demo (Cinf)

standalone Vandermonde encoder (Cinf, scale=4), shadow event #0

1 About this document

This document demonstrates the Phase-3 Vandermonde one-shot encoder running on its own, with Phase 1 and Phase 2 disabled. The Vandermonde encoder is deterministically faithful on the full rep alignment table, so it is a valid standalone encoder; this document shows exactly what it consumes (the rep atoms and the alignment table T) and what it emits.

Mode: Mode: Cinf

Scale (lambda): Scale (λ): 4

See `D0CS/phase3_vandermonde.pdf` for the mathematical derivation (DFT-node power sums, conjugate-pair compression, optional dimensional rescaling).

2 Setup

Particle types: electrons = $\{a, b, c\}$, muons = $\{p, q\}$

Group G: $G = S_{\text{electrons}} \times S_{\text{muons}} = S_3 \times S_2$, order = 12

Operations:

$|\mathbf{x}|$ (rank 1, symmetric)

$\mathbf{x} \cdot \mathbf{y}$ (rank 2, symmetric)

$\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (rank 3, antisymmetric)

Event vectors (3-D):

a: $\mathbf{a} = (-1.00, -1.00, -1.00)$

b: $\mathbf{b} = (-1.00, -1.00, 0.00)$

c: $\mathbf{c} = (1.00, -1.00, -1.00)$

p: $\mathbf{p} = (0.00, 1.00, 0.00)$

q: $\mathbf{q} = (0.00, -1.00, 0.00)$

3 Phase 3 Encoding (one-shot whole-table; rep size 25, —G—=12)

rep atoms (25): the unpruned set used by Phase 3 — every distinct atom across all G-orbits, not just one canonical representative per orbit.

[0]: [0] |a|

[1]: [1] |**b**|
 [2]: [2] |**c**|
 [3]: [3] |**p**|
 [4]: [4] |**q**|
 [5]: [5] **a** · **b**
 [6]: [6] **a** · **c**
 [7]: [7] **b** · **c**
 [8]: [8] **a** · **p**
 [9]: [9] **a** · **q**
 [10]: [10] **b** · **p**
 [11]: [11] **b** · **q**
 [12]: [12] **c** · **p**
 [13]: [13] **c** · **q**
 [14]: [14] **p** · **q**
 [15]: [15] $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$
 [16]: [16] $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$
 [17]: [17] $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$
 [18]: [18] $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$
 [19]: [19] $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$
 [20]: [20] $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$
 [21]: [21] $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$
 [22]: [22] $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$
 [23]: [23] $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$
 [24]: [24] $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$

Alignment table T (rows = rep atoms, columns = $g \in G$):

Phase 3 segments (1):

[0:3468] kind=VANDERMONDE method=vandermonde_cinf_dft len=3468: [0:3468] kind

Table 1: Phase 3 input alignment table T

Atom	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	
a	+1.7321	+1.7321	+1.7321	+1.7321	+1.4142	+1.4142	+1.4142	+1.4142	+1.7321	+1.7321
b	+1.4142	+1.4142	+1.7321	+1.7321	+1.7321	+1.7321	+1.7321	+1.7321	+1.7321	+1.7321
c	+1.7321	+1.7321	+1.4142	+1.4142	+1.7321	+1.7321	+1.7321	+1.7321	+1.4142	+1.4142
p	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000
q	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000
a · b	+2.0000	+2.0000	+1.0000	+1.0000	+2.0000	+2.0000	+0.0000	+0.0000	+1.0000	+1.0000
a · c	+1.0000	+1.0000	+2.0000	+2.0000	+0.0000	+0.0000	+2.0000	+2.0000	+0.0000	+0.0000
b · c	+0.0000	+0.0000	+0.0000	+0.0000	+1.0000	+1.0000	+1.0000	+1.0000	+2.0000	+2.0000
a · p	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000
a · q	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000
b · p	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000
b · q	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000
c · p	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000
c · q	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000
p · q	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$	-2.0000	-2.0000	+2.0000	+2.0000	+2.0000	+2.0000	-2.0000	-2.0000	-2.0000	-2.0000
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$	+1.0000	-1.0000	-2.0000	+2.0000	-1.0000	+1.0000	-1.0000	+1.0000	+2.0000	-2.0000
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$	-1.0000	+1.0000	+2.0000	-2.0000	+1.0000	-1.0000	+1.0000	-1.0000	-2.0000	+2.0000
$\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$	-2.0000	+2.0000	+1.0000	-1.0000	-1.0000	+1.0000	-1.0000	+1.0000	+1.0000	-1.0000
$\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$	+2.0000	-2.0000	-1.0000	+1.0000	+1.0000	-1.0000	+1.0000	-1.0000	-1.0000	+1.0000
$\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$	-1.0000	+1.0000	+1.0000	-1.0000	-2.0000	+2.0000	+2.0000	-2.0000	+1.0000	-1.0000
$\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$	+1.0000	-1.0000	-1.0000	+1.0000	+2.0000	-2.0000	-2.0000	+2.0000	-1.0000	+1.0000
$\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000
$\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000
$\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000

VANDERMONDE method `vandermonde_cinf_dft` length 3468

Encoded output: 3468 reals, range $[-1.8964e + 04, +1.8972e + 04]$, $\max|\cdot| = +1.8972e + 04$.

First values:

: +22.1349, +40.8414, +75.3784, +139.1609, +256.9876, +474.7126, +877.1477, +1621.2040

Last 8 values:

: +0.3352, -0.2118, +0.2561, -0.1884, +0.1961, -0.1660, +0.1497, -0.1460

4 Summary

Phase 3 encoded length: Phase 3 encoded length: 3468 reals

Total encoded length: Total encoded length: 3468 reals

repS size: $|\text{repS}| = 8$ atoms

—G—: $|G| = 12$ group elements

TeX output written to: `/Users/lester/github/Echinocoder/symcoder/demo_vandermonde.tex` Compile with: `pdflatex demo_vandermonde.tex`